

Multi-Site Company Network Design Project

Cisco Packet Tracer Team Assignment — 5-6 Students per Team

Project Overview

Your team will design and implement a realistic network for a company with two branch offices, using Cisco Packet Tracer. Each branch has different departments that must be securely segmented. Branches connect via routers and have access to local servers and shared company services. The network must use features you've learned—including VLAN, STP, NAT, DNS, DHCP, port security, ACLs, and broadcast/collision domain management.

Project Objectives

- Build a robust network topology connecting two company branches
 - Implement VLANs for department separation
 - Use STP for switch redundancy
 - Configure NAT for internet access
 - Deploy DNS and DHCP servers
 - Apply port security and ACLs for access control
 - Map and explain broadcast and collision domains
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Project Tasks

1. Requirements Gathering

- Define the company's departments in each branch (minimum: HR, IT, Sales)
- List network and security requirements for each department

2. Network Topology Design

- Draw the network diagram, labeling all devices and connections
- Specify IP addressing scheme (subnets, VLANs, private/public ranges)

3. Packet Tracer Implementation

- Deploy devices: routers, switches, servers, hubs (optional), access points (optional)
- Configure VLANs for each department
- Set up STP for switch redundancy
- Apply NAT at the router for internet access
- Configure DHCP and DNS services for each branch
- Implement port security on switches (define allowed MAC addresses per port)
- Set up ACLs on routers and switches for access control between VLANs and servers

4. Broadcast and Collision Domains

- Identify and document broadcast and collision domains in the design
- Explain the effects of segmentation and network device selection

5. Testing and Validation

- Verify inter-branch connectivity (e.g., ping, access to shared servers)
- Test VLAN separation: traffic should not pass between VLANs without proper routing/ACL
- Confirm NAT functionality: internal users can browse simulated web resources
- Demonstrate security: unauthorized devices are blocked by port security and ACLs
- Document STP operation and redundancy

6. Documentation

Prepare a report including:

- Project overview and objectives
- Network topology diagram (from Packet Tracer)
- Device and service configuration steps (summarized, with key CLI commands)
- IP addressing table
- VLAN/ACL/port security details
- Broadcast/collision domains analysis
- Testing procedures and results
- Challenges faced and solutions
- Individual contributions

7. Presentation

Create a 10-minute team presentation explaining:

- Design choices and justification

- Security measures
 - Network operation demonstration (screen recordings optional)
 - Lessons learned
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Deliverables

1. Packet Tracer file (.pkt) with complete working network
 2. Written report (PDF)
 3. Team presentation slides (PDF or PPT)
 4. A brief log of individual contributions (who did what)
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Assessment Criteria

Criterion	Description
Network Implementation	Correctness and completeness of network implementation
Security Features	VLAN, ACL, port security correctly applied
Documentation	Clear documentation and topology diagram
Team Collaboration	Effective team collaboration and division of duties
Testing	Valid testing with evidence of functioning network
Presentation	Clarity and professionalism in presentation

Table 1: Project Assessment Criteria

Hints and Tips

- Use consistent device naming (e.g., "Branch1-Switch1", "Branch2-Router")
 - Save versions and work incrementally
 - Assign configuration and testing roles to team members
 - Practice presentation before submission
 - Test each feature as you implement it—don't wait until the end
 - Document your configurations as you go
 - Use standard IP addressing schemes (e.g., 192.168.x.0/24 for private networks)
 - Keep ACL rules organized and commented
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Submission Deadline

Wednesday 10-12-2025 at 12:00 mid-night

Network Design Example Structure

Branch	Department	VLAN ID	Subnet
Branch 1	HR	10	192.168.10.0/24
Branch 1	IT	20	192.168.20.0/24
Branch 1	Sales	30	192.168.30.0/24
Branch 2	HR	40	192.168.40.0/24
Branch 2	IT	50	192.168.50.0/24
Branch 2	Sales	60	192.168.60.0/24

Table 2: Example VLAN and Subnet Allocation (Teams should design their own)

Required Network Services

Service	Location
DHCP Server	Each branch (or centralized)
DNS Server	Each branch (or centralized)
Web Server	Shared resource accessible from both branches
NAT Gateway	Branch routers

Table 3: Required Network Services

Good Luck with Your Project!

Remember: This project simulates real-world enterprise networking scenarios. Focus on understanding **why** each technology is used, not just **how** to configure it. Your ability to justify design decisions will be as important as the technical implementation.